

Nathan Berhane

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EDUCATION

Queen Mary University of London

London, England

BSc Computer Science With AI

Expected Graduation August 2029

- Worked in a team of 4 to execute in-depth research into the impact of AI in the Music industry and presented the findings in front of senior demonstrators
- Worked in a team of 4 to design a deepfake detector web extension, carried out market research and documented the user/system requirements and the design challenges
- Created a packet encoder in MIPS that encodes a domain name into a domain query packet in Big-Endian (**100%**)
- **Relevant Courses** : *Programming in Python, Programming in Java, Logic and Discrete structures, Computer Systems and Networks, Web Technology*

St Gregory's Catholic Science College

London, England

Computer Science, Maths and Furthermaths

September 2023 - July 2025

- Achieved a Gold certificate in the senior UKMT
- Results: A*A*A*

EXPERIENCE

AI and Operations Intern

London, England

Polishpad

March 2026 – April 2026

- Leveraged REST APIs to remotely update a Notion database from the backend endpoint, saving the sales team over **6 hours a week**
- Implemented a Brevo REST API to update the CRM, keeping mailing lists up-to-date
- Created a workflow to automate posting pre-written LinkedIn posts, stored on a Notion database, saving the founder **10 hours a week**
- Developed comprehensive documentation, including flow diagrams and templates, enabling the team to maintain and extend the workflows

Code Reviewer and AI trainer

London, England

Data Annotation

August 2025 – Present

- Implemented AI tools to automate checks for the compatibility of PC builds with components to train an AI model to handle customer service requests
- Completed **500+** tasks across **30+** projects, evaluated AI-generated code in line with project guidelines, improving API integration across **3+** project families
- Created atomic and self-contained rubrics to automate the evaluation of responses to a prompt
- Designed system prompts for enterprise-level APIs and created formatted (JSON/XML) user inputs and analysed output patterns to highlight weaknesses of the AI model

VOLUNTARY EXPERIENCE

Student Course Representative

London, England

Queen Mary University of London Students' Union

October 2025 – Present

- Designed engaging online surveys to collate feedback from over **400** students about the course and employed iWeaver to analyse trends and highlight common issues
- Compiled findings in a report and presented this feedback in student voice committees
- Followed up on actions after meetings by relaying important information to other students or collecting more information

PROJECTS

Context-Free Grammar Parser | *Next.js, React, Tailwind.css, Vitest, GitHub Actions* June 2026 – July 2026

- Developed a responsive web application, using the React D3 Tree component to render interactive parse trees
- Leveraged web workers to implement parallel processing when executing the CYK algorithm and CFG to CNF conversions, achieving a total blocking time of **0s**
- Automated **100+** unit tests, using a Vitest test suite and leveraged GitHub Actions and Vercel to create a **CI/CD pipeline**

UniRead | *Flask, OpenAI, base64, Docker* December 2025 – January 2026

- Developing a full-stack web application, leveraging Flask to render the web page and parse submitted files
- Implemented caching, reducing latency by **99%** in a cache hit
- Leveraged the gpt5-nano OpenAI API to generate accurate book recommendations (pre-requisites and further reading) based on a specified book
- Evaluated different OpenAI models to trade off between the cost of the API, the performance, output volume, and selected the most appropriate model based on user requirements

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, PHP

Frameworks: Next.js, React, Bootstrap, Tailwind.css, Vitest

Developer Tools: VS Code, Docker, GitHub, Git, GitHub Actions

Libraries: Flask, Tkinter, hashlib, base64, json, os, OpenAI, PyTorch, Matplotlib